# Clément Dumas

*Astra Fellow with Owain Evans, previously MATS 7 scholar with Neel Nanda.*

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## PUBLICATIONS

| lead-author papers ✦ main contributors

Training with conflicting values can induce CoT override
**C. Dumas**✦, J. Treutlein, J. Betley, O. Evans
*LessWrong 2026*

Cooperation with AIs seems to be a low-hanging fruit for better eval practices
**C. Dumas**✦
*LessWrong 2026*

Steering towards "automated grading" degrades alignment
J. Betley✦, J. Treutlein, **C. Dumas**
*LessWrong 2026*

Narrow Finetuning Leaves Clearly Readable Traces in Activation Differences
J. Minder✦, **C. Dumas**, S. Slocum, H. Casademunt, C. Holmes, R. West, N. Nanda
*ICLR 2026; NeurIPS 2025 Mechanistic Interpretability Workshop (Spotlight)*

Overcoming Sparsity Artifacts in Crosscoders to Interpret Chat-Tuning
J. Minder✦, **C. Dumas**✦, C. Juang, B. Chughtai, N. Nanda
*NeurIPS 2025*

What We Learned Trying to Diff Base and Chat Models (And Why It Matters)
**C. Dumas**✦, J. Minder✦, N. Nanda
*Alignment Forum 2025*

Separating Tongue from Thought: Activation Patching Reveals Language-Agnostic Concept Representations in Transformers
**C. Dumas**✦, C. Wendler✦, V. Veselovsky, G. Monea, R. West
*ACL 2025; ICML 2024 Mechanistic Interpretability Workshop (Spotlight)*

nnterp: A Standardized Interface for Mechanistic Interpretability of Transformers
**C. Dumas**✦
*NeurIPS 2025 Mechanistic Interpretability Workshop (Poster)*

Diff Mining: Logit Differences Reveal Finetuning Objectives
G. Kocher✦, R. West, **C. Dumas**, J. Minder
*ICLR 2026 Workshop on Principled Design for Trustworthy AI*

Pre-training Interventions, Ex Post Facto: Grafting Model Beliefs Across Checkpoints
P. Nutter✦, D. Roytburg✦, **C. Dumas**, J. Ou, S. Feng
*arXiv 2026*

Persona Cartography: Charting Language Model Personality Traits in Weight Space
L. Baines✦, A. G. Hawthorne✦, M. Koroliuk✦, I. Shalibashvili✦, **C. Dumas**, K. Voudouris, D. D. Africa
*arXiv 2026*

Believe It or Not: How Deeply do LLMs Believe Implanted Facts?
S. Slocum✦, J. Minder, **C. Dumas**, H. Sleight, R. Greenblatt, S. Marks, R. Wang
*arXiv 2025*

Activation Oracles: Training and Evaluating LLMs as General-Purpose Activation Explainers
A. Karvonen✦, J. Chua, **C. Dumas**, K. Fraser-Taliente, S. Kantamneni, J. Minder, E. Ong, A. Sen Sharma, D. Wen, O. Evans, S. Marks
*ICML 2026 (Spotlight)*

## RESEARCH POSITIONS

| | |
|---|---|
| Since Apr 2026 | **Astra Fellow**, with Owain Evans. Data attribution, then model personas |
| Sep 2025 - Jan 2026 | **SPAR mentor.** Supervised 5 junior researchers on 2 model diffing projects |
| May 2025 - Apr 2026 | **MATS extension**, Neel Nanda's stream. 1-year scholarship to continue my model diffing research |
| Winter 2025 | **MATS 7.0 scholar**, Neel Nanda's stream working on crosscoder and model diffing |
| Summer 2024 | **Research intern, EPFL**, 5 months with Robert West and Chris Wendler |
| Summer 2023 | **Research intern**, 2 months with Jobst Heitzig on Aspiration-Based Q-Learning |

## EDUCATION

| | |
|---|---|
| 2024-2025 | Research Master MVA (Mathematics, Vision, Learning), **ENS Paris-Saclay** |
| 2023-2024 | Research Master MPRI, **ENS Paris-Saclay** |
| 2022-2023 | Double Bachelor's degree in Computer Science, **ENS Paris-Saclay** |
| 2020-2022 | "Classes préparatoires" in Mathematics, Physics and CS, Lycée Thiers |

## TOOLS AND PROJECTS

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|---|---|
| tinkerscope | Chat tool for exploring and comparing weird model behaviors across checkpoints |
| nnterp | A library that standardizes access to LLM internal activations based on NNsight |
| diffing-toolkit | Implements and visualizes several model diffing techniques |
| tiny-dashboard | A small app to visualize sparse autoencoder activations |
| NDIF | Part of the team building the ecosystem behind NNsight |
| Claude plays Nomic | Multi-agent environment where Claude instances play Nomic, a game where players vote to change the rules as they play |
| CodinGame | Developed a multiplayer game; top 0.5%, 3% and 7% in bot programming contests (2021-2022) |